

China's 2023 work report and what it means: an AI post-mortem

China is taking a more nuanced approach to the overarching goal of GDP growth.

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Li Keqiang's last Two Sessions

China's so-called Two Sessions, the country's most important annual political gathering, marked a transition this year. Held in early March, it included the final work report – or summary of government business – delivered by outgoing Prime Minister, Li Keqiang, who after two five-year terms as China's prime minister, handed over to Li Qiang.

Against this backdrop, Li Keqiang's 2023 work report was blissfully short, compared to past reports. It focused mainly on reviewing 2022 economic performance and much less on the future. Still, given its symbolic importance, there is no doubt that every word in Li Keqiang's work report was measured, making it important to analyse in detail, to draw out conclusions for China's economic future.

The report (Li, 2023) did offer a few macroeconomic targets for 2023. The GDP growth target of "around 5 percent" has been read as unambitious, reflecting caution after the government missed 2022's 5.5 percent target¹. The announced employment target – 12 million new jobs – is higher than in the last five years, pointing clearly to the Chinese government's concern with rising unemployment, especially among young people. The government's short-term policy priorities are familiar as well: expanding domestic demand, modernising the industrial system, supporting public and private sector alike,

attracting more foreign investment, containing economic and financial risks, and further transitioning to a green economy. But Li Keqiang provided little detail on how these goals would be achieved.

Beyond the targets, the legacy messages Li Keqiang put into his last work report can be studied more deeply using artificial intelligence techniques to gauge the sentiment and the major ideas. For comparison purposes, we used the same techniques to also analyse the work report presented by Premier Li during the previous Two Sessions, which took place in March 2022 (Li, 2022) as China moved towards tighter mobility restrictions because of increasing COVID-19 cases.

Major topics in the 2023 work report

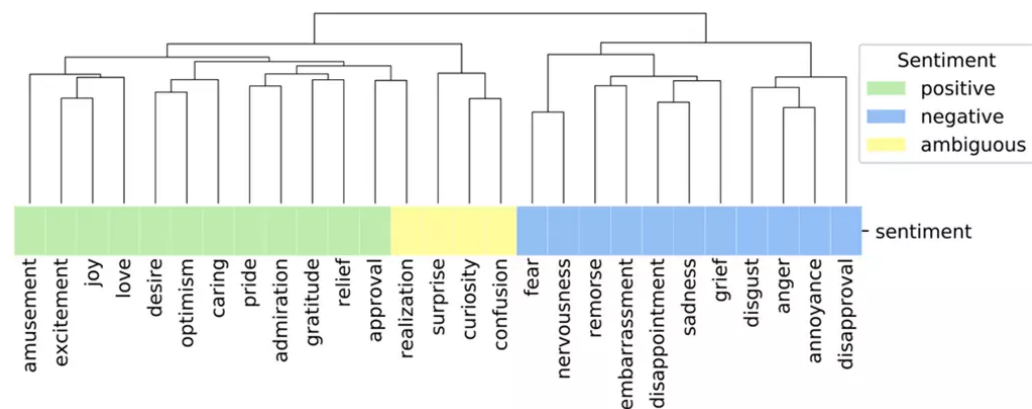
Our aim, first, was to quantify the emphasis on major topics in a comparable way. We used natural language processing to carry out a semantic classification of the 2022 and 2023 work reports, focused on five topics: GDP growth, the private sector, debt, innovation and 'other'².

This exercise showed that the greatest emphasis in 2023 is still on GDP growth, even more so than in 2022. The second most-mentioned topic was innovation, but this was mentioned less than half as much as GDP growth and, interestingly, was similarly prominent to 2022. In other words, the focus on innovation reported in the media coverage³ of the Two Sessions might be correct, but it is not the case for Li Keqiang's work report. The role of the private sector meanwhile was mentioned little in the 2023 work report – at a similar level of prominence as in 2022. Finally, the 2023 work report mentioned debt issues much more than last year, but still at a low level.

The most prominent sentiments conveyed in the work report

While words matter, such an important and highly official report tends to use very similar words year after year and the changes are very subtle. To gauge how much the tone might have changed, we identified⁴ the most frequent sentiments (Picard, 1997) expressed in Li Keqiang's 2023 work report and compared them to those in the 2022 report. To capture changes in sentiment between the two reports, we analysed each sentence separately.

Figure 2: List of sentiments used in sentence classification



Source: Alon and Ko (2021).

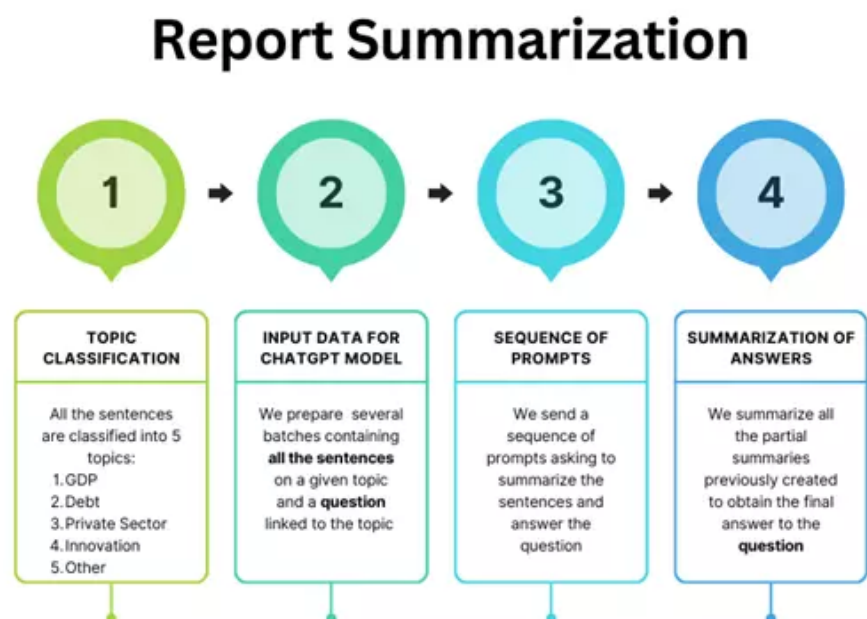
Since each work report contains fewer than 600 sentences, we focused on the top five sentiments expressed repeatedly throughout the reports. The most common sentiment is ‘neutrality’, given the official nature of the work report. However, it is interesting to note that the 2023 report has a much more official tone than the 2022 report. Meanwhile, the sentiment of ‘approval’ is much less present in 2023, and the 2022 report had a much more optimistic tone than that of 2023. Finally, while the signs of admiration and empathy (as measured by a ‘caring’ sentiment) are very limited, empathy seemed to be slightly more present in 2022, while ‘admiration’ (for example of China’s leadership or the country in general) gained some ground, starting from very low levels, in 2023.

Comparison of the main questions in the two work reports

To analyse in more detail how the 2023 work report deals with the key issues of GDP growth, the role of the private sector, debt and innovation, we summarised all the sentences related to a specific topic and answered one question (see below) based on the content of the work reports⁵. This approach is focused on information extraction from these reports using the capabilities of ChatGPT (Figure 4) and allowed us to leverage the vast general knowledge of the model to distil the major messages from the text.

In addition, we used topic modelling to compare word clouds from the 2023 report and the 2022 work report in relation to each question. To compare the reports, we selected only nouns and adjectives that are typical for each report⁶. In other words, these are not necessarily the terms that appear most, but are those that differentiate the reports from one another. This provides some insight into how the narrative changed between the two reports, and which keywords encapsulate the change.

Figure 4: Report summarisation using ChatGPT



Source: Bruegel.

Question 1: Why is the GDP target so low?

ChatGPT response: *"The GDP target is so low because the government gave priority to ensuring stable growth, employment, and prices, thus bringing about a steady economic recovery. They also adopted new ways of conducting macro regulation and avoided overreliance on investment as a means of achieving economic growth. Additionally, they took decisive measures to step up macro policy support while refraining from adopting a deluge. Despite this, China's economy is still demonstrating vast potential and momentum for further growth."*

Word cloud analysis: The word cloud for 2023 has a clear focus on the difficulties the Chinese economy faces in its recovery. The role of business is emphasised, reflecting the central government's goal of moving local governments away from relying overly on infrastructure and real-estate investment, and towards fostering higher-quality growth, securing urban employment, and increasing real incomes. The 2022 focus was similar, with a clearer emphasis on *innovation* and the implementation of the fourteenth five-year plan. Reducing the fiscal deficit to 2.8 percent was combined with an increase in expenditures to ensure stable macroeconomic conditions. This was obviously before the 2022 lockdowns.

Question 2: How much is government supporting the private sector?

ChatGPT response: *"The government supports the private sector by removing hidden barriers of various forms and ensuring all enterprises receive policy support on an equal basis. The government encourages and attracts more private capital into major state projects and projects aimed at addressing areas of weakness in order to stimulate private investment. They protect the property rights of private enterprises and the rights and interests of entrepreneurs and encourage and support the private sector and private businesses in growing and expanding, especially SMEs and self-employed individuals in business development".*

Word cloud analysis: The word cloud for 2023 associates the private sector with stimulating investment to address insufficient demand. The report also highlights unstable expectations from private investors, which will be addressed by removing hidden barriers faced by private enterprises. Private capital should also be channelled into major state projects to address areas of weakness. As for 2022, the word 'reform' appears followed closely by the word 'SOE', indicating the Chinese push to restructure Chinese state capital. The report also mentions the *"10,000 private enterprises revitalising 10,000 villages"* initiative, which emphasises the importance of private business in alleviating rural poverty.

Question 3: Is China worried about its growing debt?

ChatGPT response: *"China is taking measures to manage its growing debt, including keeping the deficit-to-GDP ratio and government debt ratio within certain limits, paying off outstanding debt from financial institution reforms, dividing fiscal powers between central and local governments, improving management of local government debt,*

establishing an individual income tax system, ensuring effective risk prevention in real-estate enterprises, improving the debt maturity mix and reducing interest payments, and preventing build-up of new debt while reducing existing debt”.

Word cloud analysis: The strong cautious stance the government now takes on debt is captured in the word cloud for the 2023 report, where the words ‘prevent’, ‘mitigation’ and ‘stable’ are mentioned often but, interestingly, together with the word ‘reform’. The ever-reoccurring mantra of high-quality growth shows up again, indicating that the move away from large fiscal deficits and reliance on the real-estate sector is seen as crucial to the rebalancing of the economy. This has been underlined by the creation of an individual income tax system. In contrast, in the 2022 work report, the word ‘debt’ is mentioned only a few times. In the 2023 report, the word ‘debt’ appears in the context of private enterprises, for which the implementation of the registration-based IPO system and the development of sound capital markets are offered as a solution. Second, major risks in the financial situation of local governments are also mentioned in this context.

Question 4: How much focus on innovation?

ChatGPT response: *“There was a significant focus on scientific and technological innovation, with breakthroughs made in various fields such as space exploration, deep sea probes, supercomputers and artificial intelligence. The capacity for innovation was enhanced continuously to sustain development. National Entrepreneurship and Innovation Week was held for eight consecutive years, with over 520 million participants. The innovation-driven development strategy was fully implemented and the industrial structure improved and upgraded. Intellectual property rights protection was bolstered and market-based mechanisms were used to encourage enterprise innovations. Annual innovation tax breaks and incentives exceeded one trillion yuan. Financial policy instruments were developed to facilitate innovation. The leading role of technological innovation was reinforced, and efforts were stepped up to secure breakthroughs in core technologies in key fields. There was a strong focus on promoting entrepreneurship and innovation nationwide, while deepening supply-side structural reform and counteracting external attempts to suppress and contain China’s development”.*

Word cloud analysis: The concepts linked to innovation in the 2023 work report include the nationwide nature of China’s innovation-driven development strategy and the necessity of financial reform. Another important concept linked to innovation is

venture capital as a channel to facilitate the growth of innovative enterprises, the principal movers in innovation. In the 2022 report, the strong connection between innovation and unbureaucratic business creation was highlighted. Innovation also appeared in the context of technology upgrading and synergies between academia, talent support and global cooperation.

Conclusions

Outgoing Chinese Premier Li Keqiang's 2023 work report is an important document in the light of this year's government reshuffling in China. Using AI-driven techniques, we have provided an in-depth analysis of this document's key messages and the sentiments behind them.

Our first takeaway is that GDP growth remains the most-mentioned topic even in the 2023 work report. However, the word-cloud analysis indicates that quality of growth is becoming even more relevant. Second, our sentiment analysis points to a much more solemn and less emotional speech in 2023, resulting in a more neutral tone than in 2022. This seems quite plausible given Li Keqiang's farewell position compared to last year when he was fully engaged, and also in the context of the huge challenge of the still unresolved COVID-19 situation and its impact on China's economy. Third, the 2023 work report also reflects a more cautious stance on debt accumulation and a more supportive environment for businesses. Innovation, which has been highlighted by all media reporting on the Two Sessions, was discussed in the context of nationwide strategic cohesion and reform of the financial system.

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Appendix

Examples of sentences for the top five sentiments identified in the 2022 and 2023 work reports:

Neutral: *"Further implementing the innovation-driven development strategy and strengthening the foundation of the real economy. We need to promote scientific and technological innovation, in order to upgrade our industries, eliminate the bottlenecks in supply and realise high-quality development through innovation."*

Approval: *"We fully and faithfully applied the new development philosophy on all fronts, worked to create a new pattern of development, promoted high-quality development and ensured both development and security."*

Optimism: *"Let us forge ahead with perseverance and resolve, promote sound and sustainable economic and social development, and keep working to build a modern socialist country in all respects."*

Caring: *"We should encourage and support the private sector and private businesses in growing and expanding, and support MSMEs and self-employed individuals in business development."*

Admiration: *“The Chinese people have pulled through with fortitude and resilience and secured a major and decisive victory in the fight against COVID-19. Throughout this truly remarkable endeavour, all localities, departments and organisations have made tremendous efforts, people in various sectors have worked together to get through tough times, and all medical personnel have remained fearless in the face of adversity.”*

Footnotes

[1] Alicia García-Herrero, Li Keqiang's farewell points to employment as China's major problem, Bruegel First Glance, 7 March 2023, <https://www.bruegel.org/first-glance/li-keqiangs-farewell-points-employment-chinas-major-problem>.

[2] To classify each sentence into one of topics, we used using few-shot learning with a GPT-Neo model. This is a method to learn new things from a few examples, similar to how people learn new tasks. We selected manually five sentences for each of the topics and provided them to the model as examples. Then, we asked the model to replicate the procedure for the entire dataset (all sentences from both reports). The approach was inspired by Saha et al (2022).

[3] See for example Zhang Tong, 'China's 'two sessions' 2023: Will science shake-up strengthen innovation or 'shock' scientists?' South China Morning Post, 11 March 2023, <https://www.scmp.com/news/china/science/article/3213086/chinas-two-sessions-2023-will-science-shake-strengthen-innovation-or-shock-scientists>, and Xinhua, 'Xi stresses high-quality development in China's modernization endeavor', 6 March 2023, <https://english.news.cn/20230306/c95a17649f8b453e9d00b3102d125da4/c.html>.

[4] Specifically, we used a RoBERTa (Liu et al, 2019) model trained on the GoEmotions dataset. See Alon and Ko (2021).

[5] We created a sequence of prompts using the sentences from the work report to force ChatGPT to use only the knowledge from these reports and summarise it into common language sentences. This approach helps contextualise the work report's sentences and obtain human-like conclusions.

[6] Selection procedure: 1). Select only sentences belonging to one of four categories: GDP, private sector, debt, innovation; 2). Find all nouns and

adjectives using nltk library; 3). Count top nouns and adjectives for report; 4). Take top n nouns and adjectives for each year and exclude those which appear in both; 5). Display the word cloud based on their frequencies.

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Endnotes

- 1 Alicia García-Herrero, Li Keqiang's farewell points to employment as China's major problem, Bruegel First Glance, 7 March 2023, <https://www.bruegel.org/first-glance/li-keqiangs-farewell-points-employ...>
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